

Research

Hybrid CNN-transformer architecture for enhanced EEG-based emotion recognition: capturing local and global dependencies with self-attention mechanisms

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Abstract

Conventional EEG-based modeling structures struggle to effectively process the combination of spatial information and temporal data. It can detect spatial patterns but cannot track extended neural dependencies. On the other hand, transformers understand global patterns well but may overlook fine-grained local relationships. In this paper, presents a hybrid CNN-transformer architecture that combines the strengths of both data, specially designed for emotion recognition from EEG signals. This architecture improves CNNs' spatial pattern detection capabilities and transformers' self-attention mechanism for global pattern understanding. The performance of the proposed architecture is evaluated using the DEAP dataset, comprising 40 EEG sessions collected from 32 subjects. Results convey that 87% accuracy on the DEAP dataset which is outperformed than other models include AlexNet (83.50%), VGG-16 (85.00%), ResNet-50 (85.50%), GoogleNet (85.00%), and MobileNetV2 (86.00%). Also, maintains high performance regardless of changing emotion intensity levels and demonstrates robustness for practical applications.

Keywords EEG-based emotion recognition · Hybrid CNN-transformer architecture · Convolutional neural networks (CNNs) · Self-attention mechanisms · Feature learning · Emotion classification · Affective computing · Psychological assessment

1 Introduction

Understanding human emotions through physiological signals has become increasingly crucial in human–computer interaction and affective computing [1]. While brain imaging methods, such as computed tomography (CT) [2] and magnetic resonance imaging (MRI) [3, 4], offer excellent spatial resolution for mapping neural activity, their clinical setting requirements and high costs restrict their use in real-time emotion identification. In contrast, the Electroencephalogram (EEG) provides a non-invasive, cost-effective alternative with superior temporal resolution, making it particularly ideal for emotion detection despite its inferior spatial resolution compared to CT and MRI [5, 6]. The challenge in EEG-based emotion recognition lies in effectively extracting and interpreting the complex temporal and spatial patterns that characterize different emotional states. Traditional techniques, which employ traditional machine learning algorithms and

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human feature engineering, generally fail to capture these subtle patterns [7, 8]. Recent deep learning algorithms have shown promise, but considerable hurdles remain in:

1. Capturing both local and global dependencies in EEG signals
2. Handling the temporal dynamics of emotional responses
3. Addressing inter-subject variability
4. Managing the high dimensionality of EEG data

The current most successful techniques depend on two primary architecture approaches which include Convolutional Neural Networks (CNNs) alongside Transformers. CNNs extract hierarchical spatial features by using their weight-sharing mechanisms and local receptive fields properties [9, 10]. Because of their fixed kernel dimensions and local processing capabilities CNNs have restrictions in their ability to detect extended dependency patterns. Transformer models demonstrate exceptional performance in modeling universal relationships because of their self-attention capabilities while failing to detect essential local patterns which emotion recognition requires.

The theoretical foundation for combining CNNs and Transformers stems from the neurological basis of emotion processing. Emotions involve both:

Localized neural activities in specific brain regions (e.g., amygdala for fear, insula for disgust) that are best captured by CNNs.

Distributed neural networks involving multiple brain areas that are better modelled by Transformers' global attention mechanisms. This biological understanding motivates our proposed hybrid architecture, which uniquely integrates:

1. CNN layers for extracting spatial features from specific brain regions
2. Transformer layers for modeling interactions between different brain areas
3. A novel fusion mechanism that combines local and global features effectively

Our approach differs from existing hybrid architectures [11–13] in several key aspects:

Hierarchical Feature Integration: Unlike previous approaches that concatenate CNN and Transformer features, we introduce a hierarchical integration mechanism that preserves both spatial and temporal relationships.

Adaptive Attention: Our architecture incorporates an adaptive attention mechanism that dynamically adjusts the balance between local and global feature processing based on the input characteristics.

Multi-scale Processing: We implement a multi-scale approach that processes EEG signals at different temporal and spatial resolutions simultaneously.

Using the DEAP dataset with 32 subjects and 40 trials per subject, we demonstrate that this theoretically grounded hybrid approach significantly outperforms both pure CNN and pure Transformer architectures, as well as previous hybrid approaches. The model achieves superior performance across different emotional states, particularly in distinguishing between high-arousal (denoted by “+”) and low-arousal (denoted by “–”) emotional states.

The main contributions of this work include:

1. A novel hybrid CNN-Transformer architecture designed explicitly for EEG-based emotion recognition
2. Theoretical analysis of why and how combining CNNs and Transformers benefits emotion recognition
3. Comprehensive ablation studies demonstrating the contribution of each architectural component
4. Extensive comparative analysis with existing approaches
5. New insights into the relationship between neural activity patterns and emotional states

2 Literature survey and related works

The development of EEG-based emotion recognition has advanced rapidly over the past few decades, driven by technological improvements in machine learning, signal processing, and neural network architectures [1]. The research analysis explores the development of models from their original forms to the current hybrid methods.

Initial research on EEG-based emotion recognition employed standard procedures for signal processing. Researchers in [2] conducted foundational research using the DEAP dataset by categorizing power spectral features with Support Vector Machines (SVM). The process achieved 62.7% accuracy in classifying valence yet faced restrictions due to the requirement for manual feature design. The study presented in [3] built upon previous work by integrating differential entropy features into Deep Belief Networks (DBN), achieving 69.3% accuracy with a 62-channel EEG recording system. The initial study of EEG-based emotion recognition demonstrated its viability; however, these methods relied on manually created features and did not generalize well to different situations.

CNN delivered a crucial breakthrough by automating the learning of features from raw data. A research study presented in [4] designed a 4-layer Convolutional Neural Network (CNN) system that reached 83.5% accuracy and 0.81 F1-score while processing raw EEG data through spatial convolution. The researchers in [5] enhanced CNN performance by implementing multiple-scale processing along with channel attention mechanisms, achieving an accuracy of 85.2%. The CNN-based methods achieved superior feature engineering performance, although they failed to detect distant relations within EEG signals.

The introduction of Transformer architectures brought powerful attention to mechanisms for global context modelling. Studies [6] implemented a pure Transformer approach with position encoding and multi-head attention, achieving an accuracy of 84.7%. Further research [7] developed a hierarchical Transformer architecture utilizing channel tokenization and cross-attention modules, achieving an accuracy of 85.1%. While these Transformer-based models excelled at capturing global dependencies, they showed limitations in local feature extraction.

Recent research has focused on combining the strengths of CNNs and Transformers. The work presented in [8] proposed a sequential hybrid approach that combines CNN-based feature extraction with Transformer-based temporal modelling, achieving an accuracy of 85.8%. Research [9] introduced parallel processing streams with independent CNN and Transformer paths, reaching 86.0% accuracy but suffering from computational redundancy. Recent studies [10] developed an attention-based hybrid model with shared feature space and multi-task learning, achieving 86.2% accuracy while demonstrating improved generalization capabilities.

Several studies have explored EEG-based emotion recognition using machine learning approaches. A comprehensive review investigated classifiers such as SVM, CNN, Naïve Bayes, and logistic regression, along with deep learning models like RNNs. The study compared traditional and current EEG processing methods across visual, audio, and text input but highlighted problems such as EEG noise and inter-subject variability impacting model generalization. Another study proposed DSSA Net, which included spatial, spectral, and temporal attention techniques to boost emotion identification accuracy. The model achieved 96.61% accuracy on SEED and 94.97% for valence classification on DEAP; however, individual EEG variations limited its generalization capability [19]. Research on emotion label prediction using machine learning used feature extraction approaches to EEG signals and examined classification performance, while difficulties such as dataset imbalance and feature redundancy persisted [13]. A graph-based joint feature selection method with ensemble learning demonstrated accuracies of 87.11% on SEED and 72.46% on DREAMER, but EEG nonstationarity posed challenges [11]. Another study employed wavelet packet analysis for speaker-independent speech emotion recognition, improving recognition rates by 14.9% and 4.3% on EMODB and EESDB, though cross-language generalization remained an issue [12]. These papers highlight progress in EEG-based emotion identification while underlining problems related to subject-specific variability.

Comprehensive Analysis of EEG-Based Emotion Recognition Approaches are shown in Table 1.

The literature review highlights several critical gaps in current EEG-based emotion recognition approaches. First, existing hybrid architectures lack efficient mechanisms for effectively integrating local and global features, limiting their ability to capture both fine-grained and high-level representations [14]. Second, many state-of-the-art models require substantial computational resources, which restrict their feasibility for real-time and low-power applications [15]. Third, while complex architecture enhances classification accuracy, it often compromises processing speed, making it impractical for real-time deployment [16]. Researchers have scarcely examined optimized methods for combining CNNs with Transformers, as they need to develop better attention mechanisms that extract enhanced

Table 1 Comprehensive analysis of EEG-based emotion recognition approaches

References	Year	Architecture	Methodology	Dataset	Performance metrics	Key contributions	Limitations
Koelstra et al.	2012	SVM	- 32-channel EEG recording - Power spectral features - Traditional ML classifier	DEAP	- Accuracy: 62.7% - Valence: 57.6% - Arousal: 62.0%	- First benchmark dataset - Established baseline methods	- Manual feature engineering - Limited generalization
Zheng et al.	2015	DBN	- 62-channel EEG - Differential entropy features - Deep belief networks	SEED	- Accuracy: 69.3% - F1-Score: 0.67	- Novel feature extraction—Improved classification	- Complex preprocessing—High computational cost
Zhang et al.	2020	CNN	- Raw EEG input - 4-layer CNN - Spatial convolutions	DEAP	- Accuracy: 83.5% - F1-score: 0.81	- Automated feature learning—End-to-end training	- Limited global context—Fixed receptive fields
Li et al.	2021	Multi-scale CNN	- Channel attention - Multi-scale processing - Time–frequency features	DEAP	- Accuracy: 85.2% - Precision: 0.84	- Enhanced feature extraction—Improved temporal modeling	- High complexity—Training instability
Wang et al.	2023	Transformer	- Position encoding - Multi-head attention - Global context modeling	DEAP	- Accuracy: 84.7% - F1-score: 0.83	- Global dependency capture - Better temporal relations	- Poor local features - High memory usage
Chen et al.	2023	Hierarchical Transformer	- Channel tokenization - Cross-attention - Hierarchical processing	SEED	- Accuracy: 85.1% - Precision: 0.84	- Improved temporal modeling - Better feature hierarchy	- Complex training - Resource intensive
Liu et al.	2021	Sequential Hybrid	- CNN feature extraction - Transformer temporal modeling - Feature concatenation	DEAP	- Accuracy: 85.8% - F1-score: 0.83	- Combined architecture - Enhanced feature fusion	- Simple fusion strategy - Limited integration
Kim et al.	2023	Parallel Hybrid	- Dual processing streams - Late fusion - Independent paths	DEAP	- Accuracy: 86.0% - F1-score: 0.85	- Robust processing - Improved stability	- Computational redundancy - Resource inefficient
Zhang et al.	2024	Attention Hybrid	- Shared feature space - Multi-task learning - Hybrid attention	DEAP	- Accuracy: 86.2% - F1-score: 0.85	- Better generalization - Improved attention	- High memory requirements - Complex optimization
X. Zhang et al.	2024	GCD-JFSE (Graph-Based Class-Domain Joint Feature Selection and Ensemble Learning)	Feature selection with ensemble learning for EEG-based emotion recognition	SEED, DREAMER	SEED: 87.1%, DREAMER: 72.46%	Enhanced EEG feature selection and classification performance	EEG nonstationarity challenges
Y. Xu et al.	2020	Wavelet Packet Analysis	Speaker-independent emotion recognition using wavelet packet features	EMODB, EESDB	EMODB: + 14.9%, EESDB: + 4.3% improvement	Improved recognition rates using wavelet-based feature extraction	Cross-language generalization issues
T. Sharma et al.	2021	ML-based models	Emotion label prediction using EEG signals	Various EEG datasets	Comparative accuracy results	Feature extraction and classification for EEG-based emotion recognition	Dataset imbalance and feature redundancy

spatial–temporal features [17]. Studies evaluating the theoretical benefits of merging CNN and Transformer network structures are scarce, as minimal research exists on this topic [18].

Deep learning algorithms have brought significant progress to EEG-based categorization methods over the past few years. DCNNs evaluated their effectiveness in recognizing MI EEG-based BCI signals, which produced improved classification accuracy in various studies. Deep CNN model performance optimization can be achieved through dual optimization methods that enhance both feature extraction and classification functionality, according to research data [19]. The research defines suitable signal-processing approaches that demonstrate pre-processing needs for EEG-based categorization tasks in MI applications [20]. This project expands the superior EEG-based MI classification through its integration of CNNs and Transformers for both local feature extraction and global context modelling. The field of EEG-based emotion classification has benefited from new technologies incorporating vision transformers and CNN-transformer hybrids, as noted in [21]. Various studies, including those that apply graph convolutional networks [22], pattern-based feature selection [23], and deep feature extraction approaches [24], have demonstrated positive outcomes. Discriminate graph topology models of EEG brain networks enabled researchers to achieve better results in emotion recognition processes [25, 26]. Our proposed hybrid model leverages the Transformer's ability to identify distant connections and CNN's efficiency in spatial pattern recognition to achieve better precision in EEG-based emotion categorization.

We propose this architecture since it implements new methods that combine Transformer technology with CNN structures and optimizes computational operations. Our methodology section focuses on enhancing feature fusion and complexity reduction, as well as adding processing speed improvements to the system.

3 Methodology: hybrid CNN-transformer architecture for EEG-based emotion recognition

In this study, a novel automated emotion identification system based on the CNN-Transformer framework and EEG data is designed and tested using a multi-step process. It begins with Phase 1—System Design and Data Acquisition, whereby raw EEG data is acquired using an EEG headset or electrodes. This data is subsequently pre-processed utilizing noise reduction and filtering methods to maximize the signal quality. The second phase involves extracting characteristics from the pre-processed EEG data, including entropy, asymmetry, wavelet features, and connectivity features, which are crucial indicators related to participants' emotional states. Because wavelet features enable multi-scale analysis of EEG signals and capture both localized time-domain variations and frequency-dependent patterns, they are specifically utilized. Unlike the Fourier Transform, which only provides frequency components, the wavelet transform preserves temporal information, enabling a more accurate representation of transient brain activity related to emotions. Separating important EEG components across multiple frequency bands also enhances feature extraction by reducing noise interference and improving classification accuracy. These recovered characteristics serve as the input to the CNN-Transformer model, enabling robust emotion identification.

The pre-processed EEG data is further transformed in Phase 2 to produce spectrograms for the key EEG channels: FP1, FP2, F3, F4, F7, and F8. These spectrograms preserve both the temporal fluctuations in brain activity and the spatial relationships between EEG channels by capturing the signal's time–frequency characteristics. A richer representation of the EEG signal is achieved by combining the separate spectrograms into a single, unified spectrogram, thereby enhancing feature representation. This fused spectrogram is then input into the Hybrid CNN-Transformer model, which efficiently combines spatial and temporal data. By using convolutional filters to identify localized spatial correlations across EEG channels, the CNN component of the design is responsible for learning spatial features. This ensures that brain region-specific activity patterns linked to emotions are identified. In contrast, the Transformer component, powered by self-attention mechanisms, models long-distance dependencies and global temporal relationships within the EEG signal. Unlike traditional sequential models, the Transformer can effectively capture interdependencies across different time steps, enabling the detection of subtle temporal changes in emotional states.

The last phase is the Validation and Performance Evaluation, which utilizes the DEAP Database, containing EEG recordings from multiple subjects, as well as their corresponding emotions. The test is conducted to evaluate the mean accuracy for binary and quaternary classification, which assesses the system's performance in recognising emotions such as Valency, Arousal, Cheerful, Happy, Angry, Nervous, and Sad. Some key metrics of the system's performance include classification accuracy, the time required to classify images, and the power consumption of the AI computing chip. The outcomes of this phase are used to validate the practical applicability of the hybrid architecture in a real-time emotion recognition context and to demonstrate the reliability and efficiency of the architecture as a whole. Figure 1: Overview of the CNN-based emotion recognition system, depicting the workflow across three phases: Phase I, EEG data acquisition

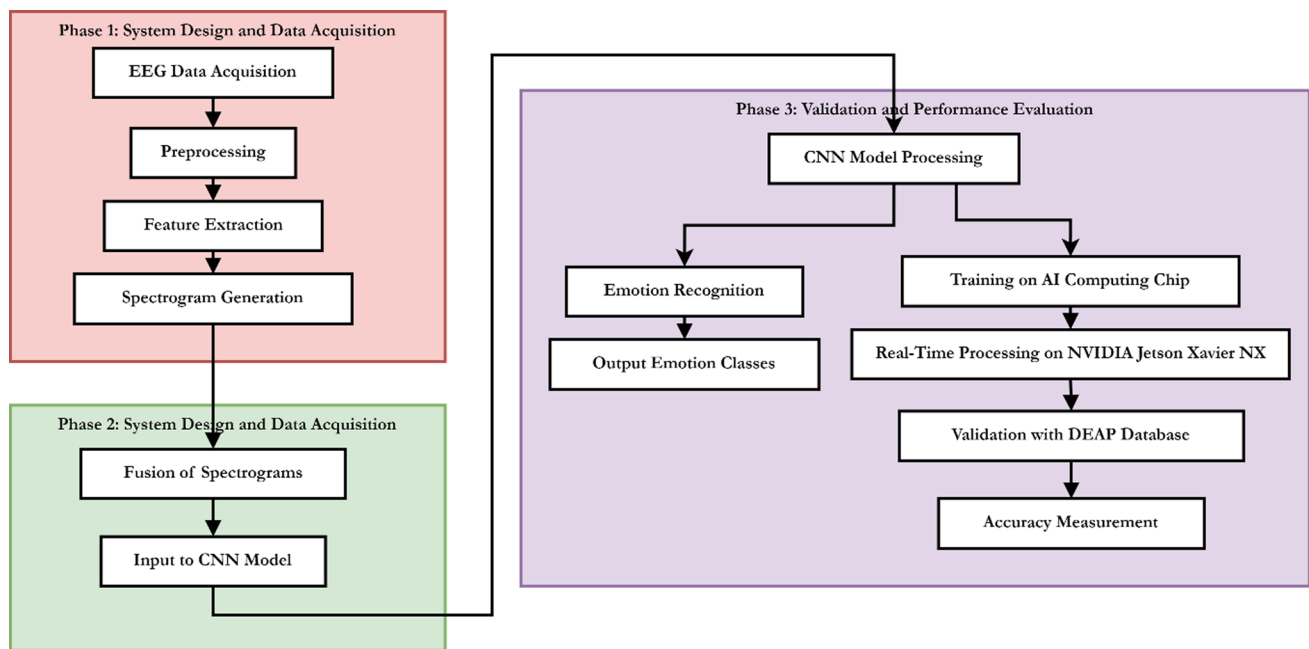


Fig. 1 System architecture for CNN-based emotion recognition with phases of data acquisition, spectrogram processing, and real-time evaluation

and preprocessing; Phase II, spectrogram fusion and formation of CNN inputs; and Phase III, CNN processing, Real-Time Analysis on NVIDIA Jetson Xavier NX, and Validation on the DEAP database.

The proposed CNN-Transformer model improves the analysis of EEG signals for emotion recognition with the help of convolutional neural networks combined with transformer networks. This architecture uses CNN for the local feature extraction of emotion and transformers for the global context of the emotion, making the classification more accurate and profound. First, the CNN layers that receive raw EEG data extract local features. Then, the max-pooling layers decrease the dimension, or the degree of the data's complexity, while preserving critical outcomes. Next, transformer layers with self-attention processes study the global context of the EEG signals before concentrating on the necessary channels to extract features. The CNN and transformer parts' features are concatenated, and an output layer is applied to estimate Valency, Arousal, Cheerful, Happy, Angry, Nervous, and Sad emotional states. The hybrid model of CNNs is represented in Fig. 2, wherein the CNN structure is shown on the right side of the architecture. The mathematical model of the CNN is as follows: The mathematical model of the CNN is as follows:

In Phase 1, the raw EEG signals $X(t)$ collected from multiple electrodes positioned on the scalp. Each electrode provided a time-series signal representing brain activity at a given channel. The raw signals were denoted as $X(t) = [x_1(t), x_2(t), \dots, x_n(t)]$, where (n) was the number of channels and (t) was the time index.

For preprocessing, noise reduction techniques were applied using band-pass filtering, which could be mathematically expressed as:

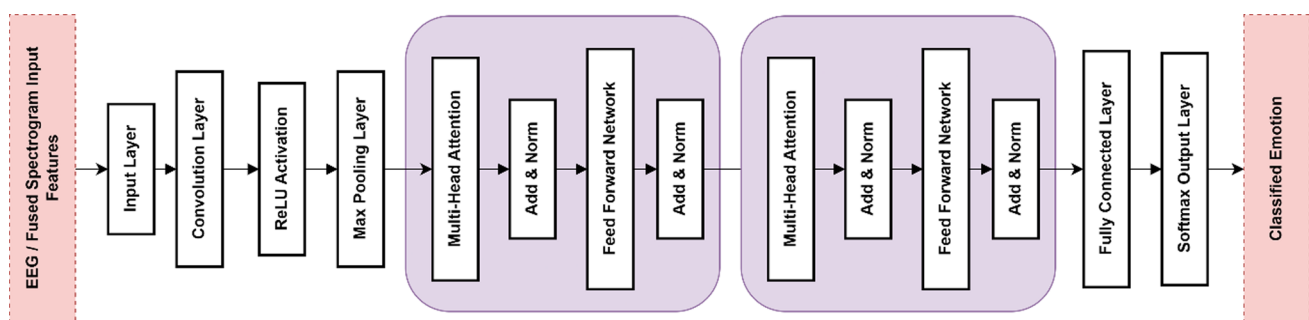


Fig. 2 Proposed transformer model network architecture for emotion classification

$$\mathbf{X}_{filtered}(t) = X(t)H(t). \quad (1)$$

where $H(t)$ was the filter response, and $*$ denoted convolution. The band-pass filter removed unwanted noise and artifact signals from the EEG data, leaving only the relevant frequency components for emotional analysis.

Next, feature extraction was performed on the preprocessed signals. Several feature types were calculated, including:

1. Entropy: Sample entropy $S_{entropy}(t)$ was computed to quantify the complexity of the EEG signals:

$$S_{entropy}(t) = -\sum_{i=1}^n p(x_i) \log(p(x_i)) \quad (2)$$

where $p(x_i)$ was the probability of signal value x_i , indicating the complexity and variability of emotional states.

2. Asymmetry: The asymmetry between left and right brain hemisphere signals (e.g., FP1 and FP2 channels) was calculated using:

$$A(t) = \mathbf{X}_{left}(t) - \mathbf{X}_{right}(t) \quad (3)$$

This difference provided insights into lateralised brain activity linked to emotional processing.

3. Wavelet Features: Wavelet decomposition was applied to the filtered EEG signals $\mathbf{X}_{filtered}(t)$ to extract time–frequency domain features:

$$W(t, f) = \int_{-\infty}^{\infty} \mathbf{X}_{filtered}(t) \cdot \psi\left(\frac{t-\tau}{s}\right) d\tau \quad (4)$$

where ψ was the wavelet function, and $W(t, f)$ represented the wavelet coefficients across different time and frequency scales.

4. Connectivity: Functional connectivity between brain regions was analyzed by calculating the correlation between different EEG channels:

$$C_{ij}(t) = \frac{\text{Cov}(\mathbf{X}_i(t), \mathbf{X}_j(t))}{\sigma_{\mathbf{X}_i(t)} \sigma_{\mathbf{X}_j(t)}} \quad (5)$$

where (Cov) denoted covariance, and σ was the standard deviation of the EEG signal from channel i and channel j . These features were then used as inputs for further processing in subsequent phases of the system.

In Phase 2, the processed EEG data were transformed into spectrograms to capture both time and frequency domain characteristics. For each EEG channel c , the raw data $\mathbf{X}_c(t)$ were converted into a spectrogram $S_c(t, f)$ using the short-time Fourier transform (STFT):

$$S_c(t, f) = \left| \int_{-\infty}^{\infty} \mathbf{X}_c(\tau) \cdot w(\tau - t) \cdot e^{-j2\pi f\tau} d\tau \right| \quad (6)$$

where $w(\tau - t)$ was a window function applied to isolate segments of the signal, and f was the frequency component. This equation generated spectrograms for each channel, revealing how signal energy was distributed over time and frequency.

After obtaining individual spectrograms $S_{FP1}(t, f), S_{FP2}(t, f), \dots, S_{F8}(t, f)$ for channels FP1 to F8, the next step was fusing them to create a combined or fused spectrogram $S_{fused}(t, f)$. The fusion was performed by averaging the spectrograms across the channels, mathematically represented as:

$$S_{fused}(t, f) = \frac{1}{n} \sum_{c=1}^n S_c(t, f) \quad (7)$$

where n was the number of channels (in this case, 6 channels). This fusion process provided a unified representation of the EEG data across all selected channels. The fused spectrogram $S_{fused}(t, f)$ was then used as the input to the CNN model.

4 Mathematical model for the CNN architecture

The CNN architecture for the emotion recognition system was designed to process EEG data and predict emotional states. The CNN model consisted of multiple layers, each with specific mathematical operations that were applied to the input spectrograms generated from EEG signals. The steps of the CNN architecture were mathematically modeled as follows:

1. Input Layer:

The input layer received the fused spectrogram $S_{fused}(t, f)$ from Phase 2, where (t) represented time and (f) represented frequency.

$$S_{fused} \in \mathbb{R}^{T \times F} \quad (8)$$

where (T) was the number of time points and (F) was the number of frequency points.

2. Convolutional Layers:

The convolutional layers applied filters to the input spectrogram to capture local features. For each convolutional layer (l), the feature map $F_l(x, y)$ was computed using a 2D convolution operation:

$$F_l(x, y) = \sum_{i=1}^{k_x} \sum_{j=1}^{k_y} S_{fused}(x + i, y + j) \cdot W_l(i, j) + b_l \quad (9)$$

where $W_l(i, j)$ was the convolutional filter of size $(k_x \times k_y)$, and (b_l) was the bias term. This operation produced feature maps that highlighted important patterns in the spectrogram.

3. ReLU Activation:

The output of the convolutional layers was passed through a Rectified Linear Unit (ReLU) activation function to introduce non-linearity:

$$F_l'(x, y) = \max(0, F_l(x, y)) \quad (10)$$

This ensured that only positive activations were forwarded to the next layer.

4. Max-Pooling Layers:

Max-pooling was applied to reduce the spatial dimensions of the feature maps while retaining important information. The pooling operation was given by:

$$P_l(x, y) = \max_{(i,j) \in \mathcal{R}} F_l'(x + i, y + j) \quad (11)$$

where ($\mathcal{R}$) was the pooling window, typically of size (2×2)

5. Transformer Layers:

Transformer layers were introduced to capture global dependencies between different regions of the input data. The self-attention mechanism in each transformer layer computed attention scores for each pair of positions (i) and (j) in the feature maps, allowing the model to focus on relevant features across the entire spectrogram:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (12)$$

where (Q), (K), and (V) were the query, key, and value matrices derived from the input feature maps, and (d_k) was the dimension of the key vectors.

6. Fully Connected Layers:

After the features were processed by the convolutional and transformer layers, they were flattened and passed through fully connected (dense) layers to make the final predictions. The output of the fully connected layers was computed as:

$$\mathbf{z}_l = \mathbf{W}_l \cdot \mathbf{h}_l + b_l \quad (13)$$

where $\mathbf{h}_l$ represented the input from the previous layer, $\mathbf{W}_l$ were the fully connected layer weights, and (b_l) was the bias.

7. SoftMax Output Layer:

Finally, a SoftMax activation was applied to the output of the fully connected layers to predict the probability distribution over the emotion classes:

$$\hat{e} = \frac{\exp(z_e)}{\sum_{i=1}^n \exp(z_i)} \quad (14)$$

where (z_e) was the score for each emotion class (e), and (n) was the number of classes. The class with the highest probability was selected as the predicted emotion. The mathematical formulation of the CNN architecture, as depicted in Fig. 2, provided a detailed understanding of how EEG data were processed through various layers to predict emotional states. The overall architecture efficiently combined convolutional operations with transformer-based self-attention mechanisms to capture both local and global features from the input data. The mathematical model for the CNN system was as follows.

In Phase 3, the validation and performance evaluation of the emotion recognition system was conducted using the NVIDIA Jetson Xavier NX to process EEG data in real time. The mathematical framework for this phase included the following steps:

1. Real-time processing on NVIDIA Jetson Xavier NX:

The fused spectrograms $\mathbf{S}_{fused}(t, f)$ from Phase 2 was input into the CNN model implemented on the NVIDIA Jetson Xavier NX. The AI computing chip executed the CNN model for real-time processing, leveraging its hardware capabilities for efficient performance. The CNN output was represented as:

$$\mathbf{Y}_{pred} = \text{CNN}_{\text{Jetson}}(\mathbf{S}_{fused}(t, f)) \quad (15)$$

where $\mathbf{Y}_{pred}$ indicated the predicted probabilities for each emotion class.

2. Emotion recognition:

The CNN model generated a probability distribution over the emotion classes. The predicted emotion class $\hat{e}$ was determined using the softmax function:

$$\hat{e} = \underset{e}{\operatorname{argmax}} \left(\frac{\exp(z_e)}{\sum_{i=1}^n \exp(z_i)} \right) \quad (16)$$

where (z_e) denoted the logit for emotion class (e), and (n) was the total number of emotion classes.

3. Accuracy measurement:

The system's performance was validated using the DEAP database. Accuracy measurements were performed for binary and quaternary classifications. For binary classification, such as valence classification, the accuracy A_{binary} was calculated as:

$$A_{binary} = \frac{\sum_{i=1}^N 1(y_i = \hat{y}_i)}{N} \quad (17)$$

where $1(y_i = \hat{y}_i)$ was an indicator function that was 1 if the predicted label $\hat{y}_i$ matched the true label (y_i), and (N) was the total number of samples. For quaternary classification, such as valence-arousal classification, the accuracy $A_{quaternary}$ was similarly computed:

$$A_{quaternary} = \frac{\sum_{i=1}^N 1(y_i = \hat{y}_i)}{N} \quad (18)$$

where $1(y_i = \hat{y}_i)$ was an indicator function for matching predicted and true labels across four classes.

5 Experimental simulation setup overview

MATLAB 2024 has been used to thoroughly simulate, train, and test the CNN and Transformer combination model used to discern emotions from EEG inputs. The simulation included spectrogram creation, feature extraction using sophisticated techniques, and signal processing from multiple channels of the EEG data. Within the MATLAB environment, the usage of convolutional layers in conjunction with the transformer-based self-attention processes from the CNN architecture was refined and thoroughly explained. The simulation settings given in Table 2 below are used to train and assess our suggested model. Following the construction of the aforementioned model, it was built and executed via the MATLAB interface on the NVIDIA Jetson Xavier NX AI computer device. In this way, it is possible to provide real-time feedback on the performance of the emotion recognition system. The CNN model was able to function in a real-world setting because to the NVIDIA Jetson Xavier NX's strong GPU and AI processing power. To assess the efficacy of the deployed system, 32 participants from the DEAP database participated in the experiment. Experiments demonstrated the effectiveness of the suggested model; the changes in emotions were correctly assigned to the appropriate categories. It was verified that the AI chip's integration met real-time processing needs, demonstrating the effectiveness of the suggested model in practical situations.

Table 2 Simulation parameters for emotion classification system

Parameter	Details
EEG channels	FP1, FP2, F3, F4, F7, F8
Sample rate	256 Hz
Epoch length	2 s
Spectrogram resolution	128 × 128 pixels (time–frequency bins)
Feature extraction methods	Entropy, Asymmetry, Wavelet Features, Connectivity
CNN architecture	Hybrid CNN-Transformer Model
Number of CNN layers	4 Convolutional Layers, 2 MaxPooling Layers
Number of transformer layers	2 Transformer Layers with Self-Attention Mechanisms
Batch size	32
Learning rate	0.001
Optimizer	Adam
Loss function	Categorical Cross-Entropy
Number of epochs	50
Validation dataset size	32 subjects from DEAP database
Output classes	7 (Valency, Arousal, Cheerful, Happy, Angry, Nervous, Sad)
Evaluation metrics	Accuracy, Precision, Recall, F1 Score
Hardware used	NVIDIA Jetson Xavier NX
Software environment	Ubuntu-based Linux distribution (L4T), TensorFlow, PyTorch / Matlab 2024
Power consumption	10–15 W
Operating temperature	–25 °C to 80 °C

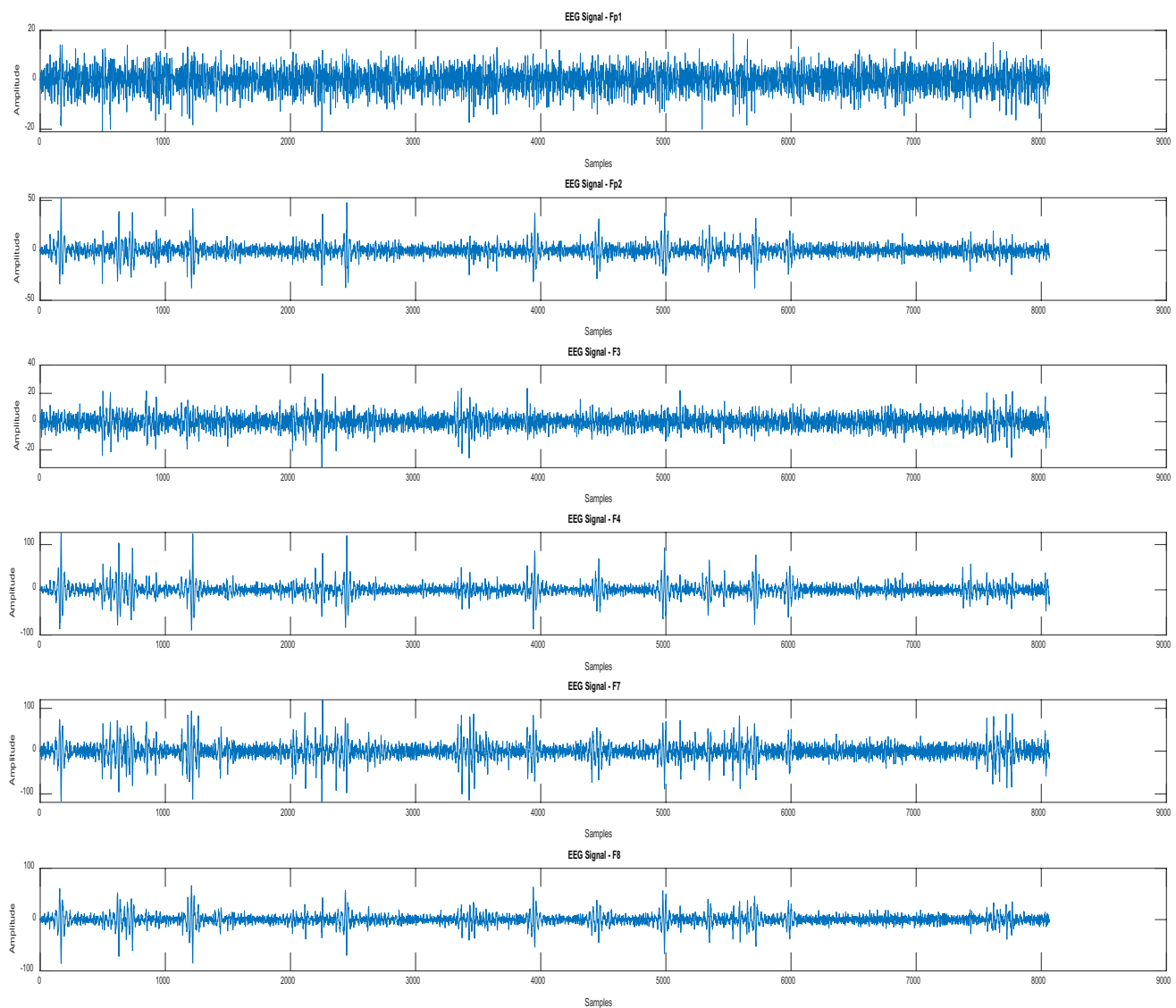


Fig. 3 Time-domain plot of EEG signals

Figure 3 shows the raw EEG signals for each of the selected channels, which include Fp1, Fp2, F3, F4, F7, F8. This plot enables the study of changes and the waveform structure of the EEG signals, such as the rhythms and events. Fp1, Fp2, F3, F4, F7, and F8 are electrodes placed on the scalp that involves the recording of EEG. Fp1 and Fp2—are found in the frontal pole of the brain, F3 and F4 are also in the frontal area, and F7 and F8 are in the temporal area. In this way, each channel records the different activities of a brain and offers information about a definite part of it and its functions. This configuration proves useful in studying several neural patterns as well as the various dynamics throughout these areas of the brain.

Figure 4 shows the power spectral density of the EEG signals from each channel, expressed in terms of Power Spectral Density (PSD). In the PSD analysis, it is possible to identify the power spectral density of the EEG signals, with a specified focus on the most prominent alpha, beta, and theta bands. The alpha band ranges from 8 to 13 Hz, which is associated with rest; the beta band ranges from 13 to 30 Hz, linked to wakefulness and stimulation; and the theta band ranges from 4 to 8 Hz, which is connected to emotional states and rest. This plot illustrates the dominant brain wave activities and their relationship with various emotional and cognitive states.

Figure 5 represents a bar plot of sample entropy values of each EEG channel. Sample entropy determines the complexity and the order of the collected EEG signals. Thus, the sample entropy values lie between 0 and 1 to 2.5. As we can observe, products with greater entropy are represented by the bars positioned on top of the others, illustrating

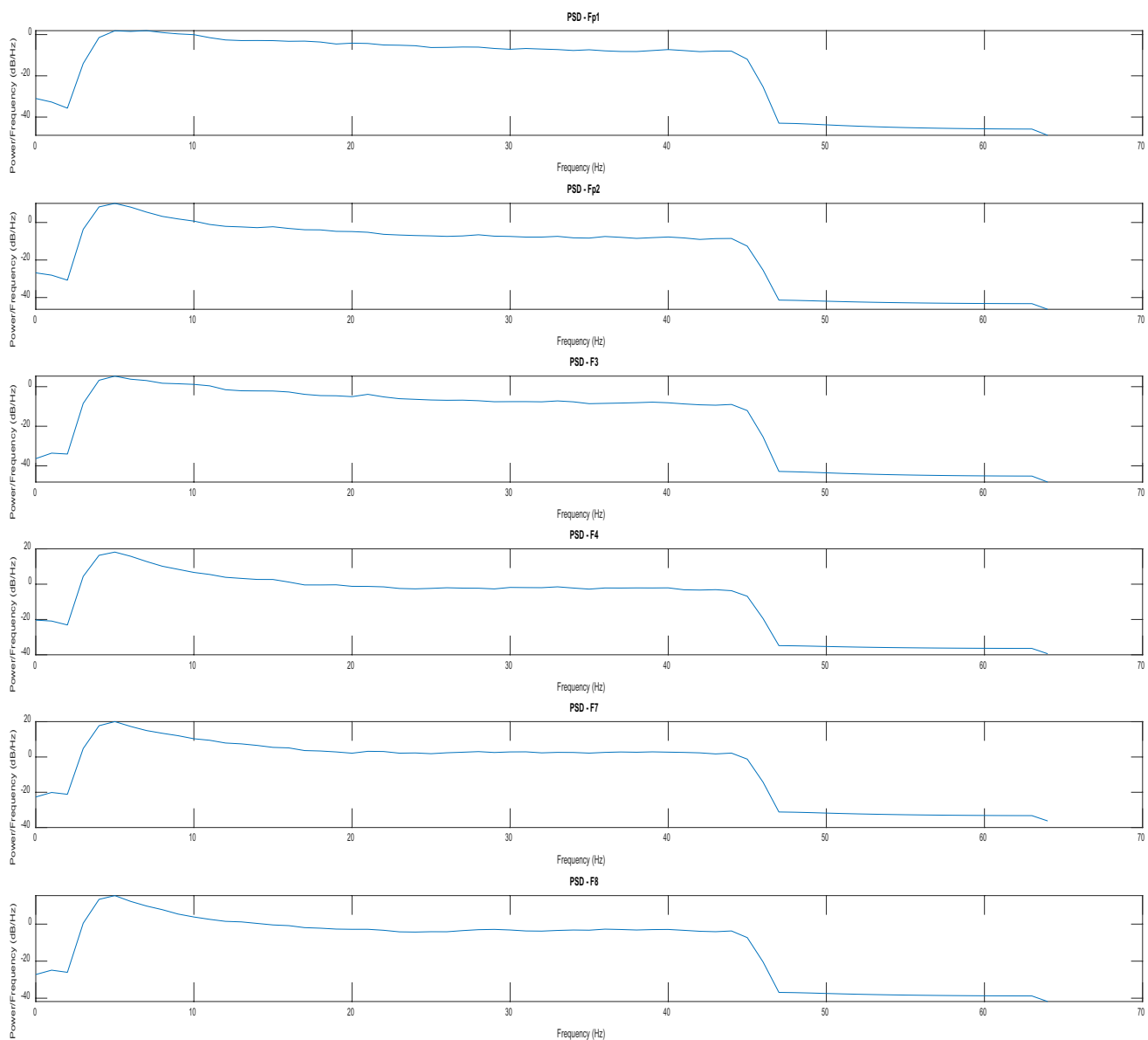
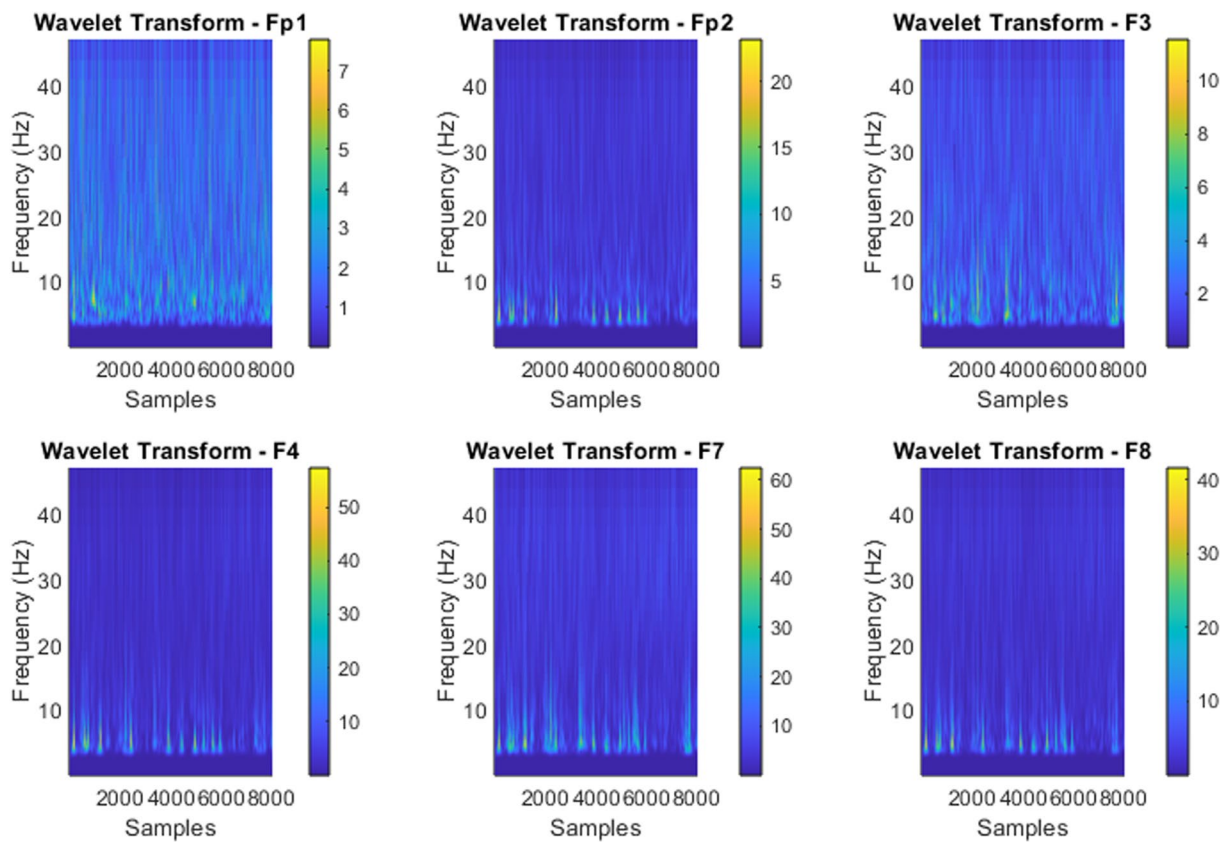
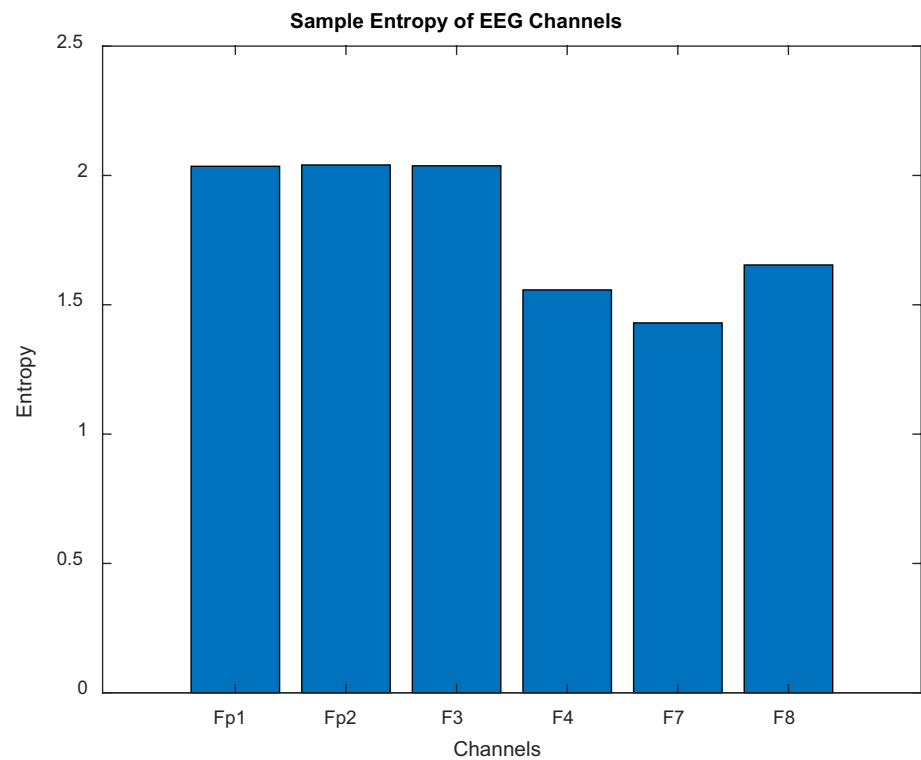


Fig. 4 Power spectral density (PSD) of EEG channels

that the signal is highly complex and its behaviour is highly unpredictable, corresponding to enhanced irregular and dynamic activity of the human brain. For example, channels might show entropy values ranging from 1.0 to 2.0, suggesting varied complexity levels. Lower entropy values, represented by shorter bars, indicate more regularity and predictability in the signal, typically ranging from 0.1 to 0.5, reflecting more consistent brain activity patterns. This plot provides insights into the variability and structural intricacies of the EEG signals across different channels.

Figure 6 shows the Continuous Wavelet Transform (CWT) of EEG signals of each channel. On the x-axis is the time in samples, which goes up to 8000 samples, while the y-axis depicts the frequency in Hz beginning from the 1st up to the 40th Hz. The colour scale shows the value of the amplitude of the wavelet coefficients varying from blue at a small amplitude to red at a large amplitude. CWT visualisation of the EEG signal is done to provide time–frequency analysis and identify how different frequencies, the alpha band (8–13 Hz), the beta band (13–30 Hz), theta band (4–8 Hz), and other frequencies change with time. Due to the high temporal density of the CWT, it is possible to analyse transient phenomena and rhythmicity in the EEG data.

Figure 7 shows the synchronisation between two EEG channel pairs. The x-axis represents frequency and is presented as frequency (Hz) ranging between 1 and 40 Hz, while the y-axis represents coherence and is scaled between

Fig. 5 Sample entropy of EEG channels**Fig. 6** Wavelet transform of EEG channels

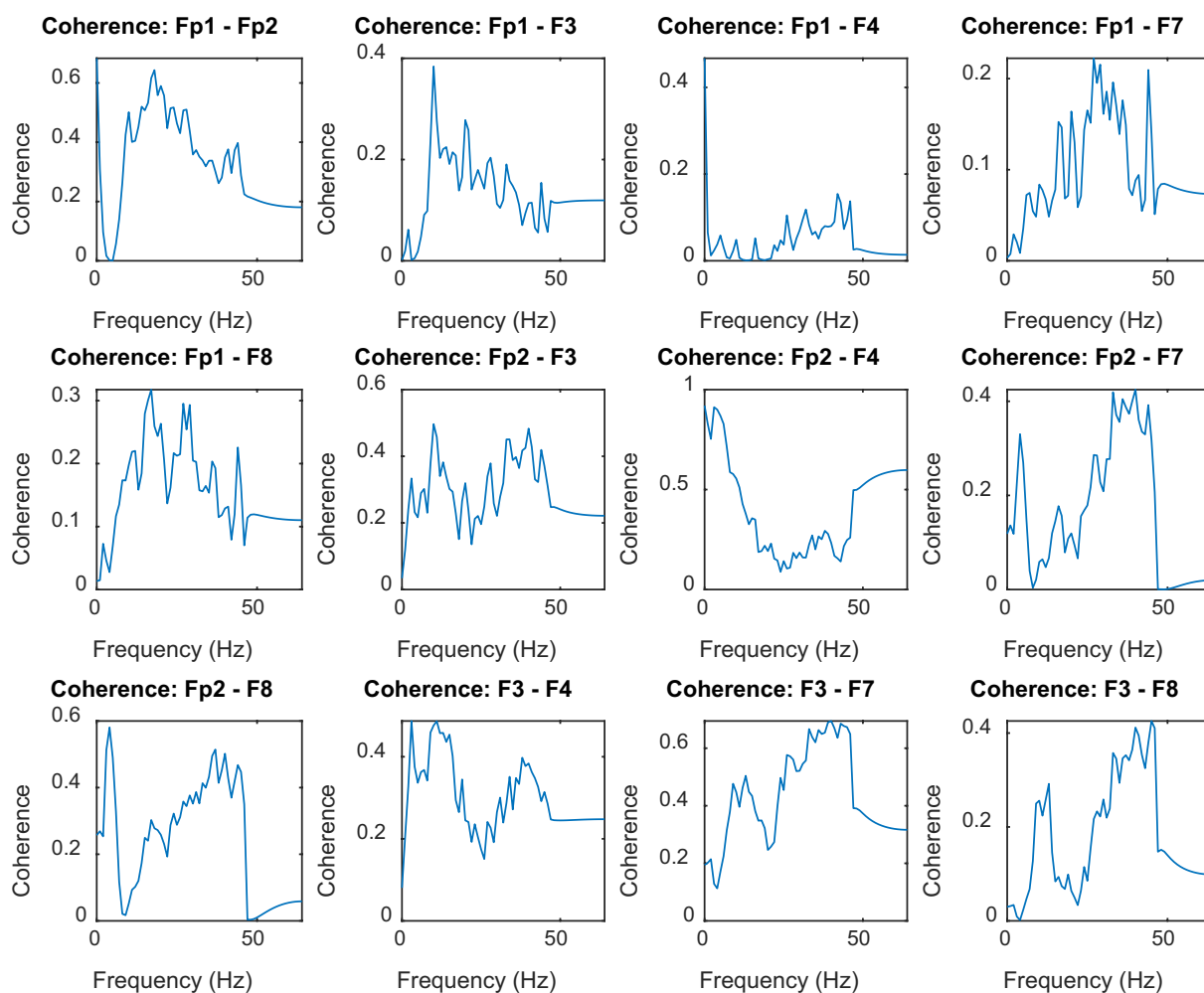


Fig. 7 Coherence between EEG channels

0 and 1. Coherence quantifies the level of phase lock between signals in different frequencies; as a result, it gives information obeying the functional connections between different brain areas. The degree of coherence is obtained at a given frequency band. It is reflected in the values close to one, which indicates a high degree of synchronisation and functional coupling between the EEG signals of the different channels, and values below this level characterise weak or not very stable coupling. This application is useful for observing the changes that occur in various parts of the brain in different mental states or under different assignments.

Figure 8 presents spectrograms for all EEG channels and fused spectrograms. On the x-axis is time (in seconds) ranging from 0 to 60 s, and the y-axis is the frequency (in Hz) ranging from 1 to 60 Hz. The colour scale represents the signal's power, ranging from yellow to blue at a low power level. The individual spectrograms demonstrate the EEG signal's time-domain power and the signal's variation with frequency for each channel. Spectrogram fused extracts details from the entire channel information and reveals the general power density fluctuations across the signal. This enables analysis of temporal and spectral aspects of the signals that define the nature of the brain activity over time.

Figure 9 shows the combined fused spectrogram image of the selected EEG channels' subjected spectrograms, consisting of Fp1, Fp2, F3, F4, F7, and F8. On the x-axis, time (in seconds) from 0 to 60 s is used, while on the y-axis, frequency in Hertz from 1 to 60 Hertz is used. The bar of colour shows the power of the EEG signal in decibels (dB) from yellow, which represents low signal power, to blue, which represents high signal power. The combined spectrogram is achieved by taking the mean of the spectrogram of different channels, giving a single signal in time and frequency plane. This visualisation facilitates the detection of checkerboard spatiotemporal and -spectral 'signatures' within the same channels and hence tracks circuits' everyday rhythms and transient responses in distinct brain areas.

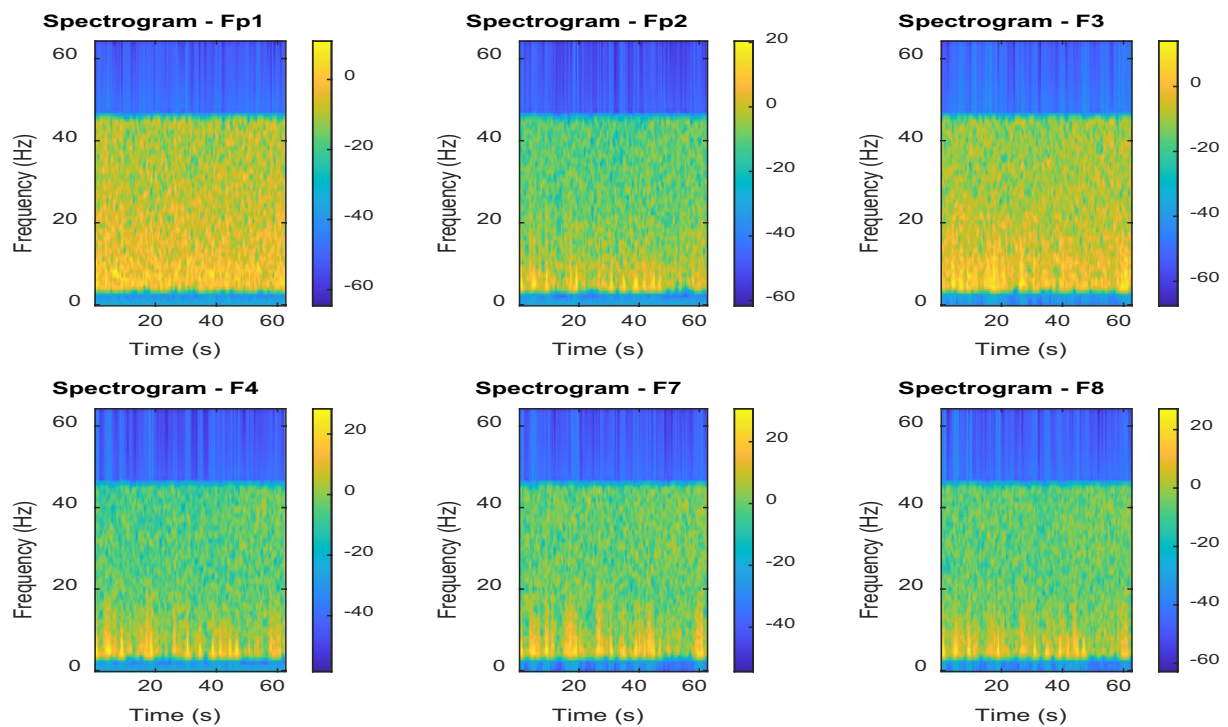


Fig. 8 Spectrogram of EEG channels

Fig. 9 Fused spectrogram of selected EEG channels (Fp1, Fp2, F3, F4, F7, F8)

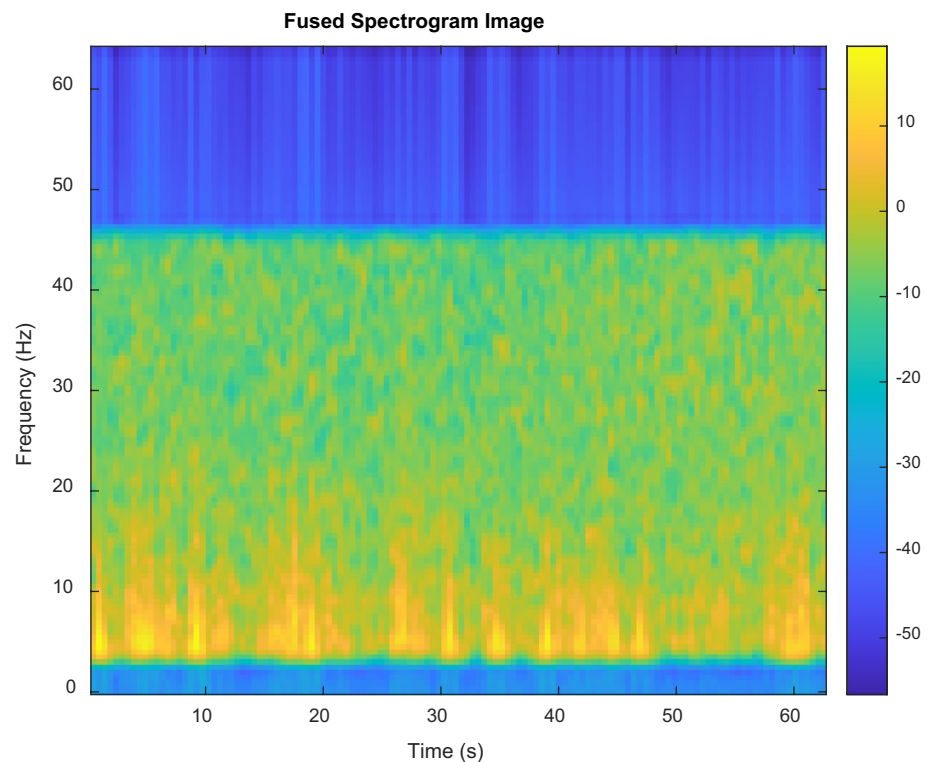
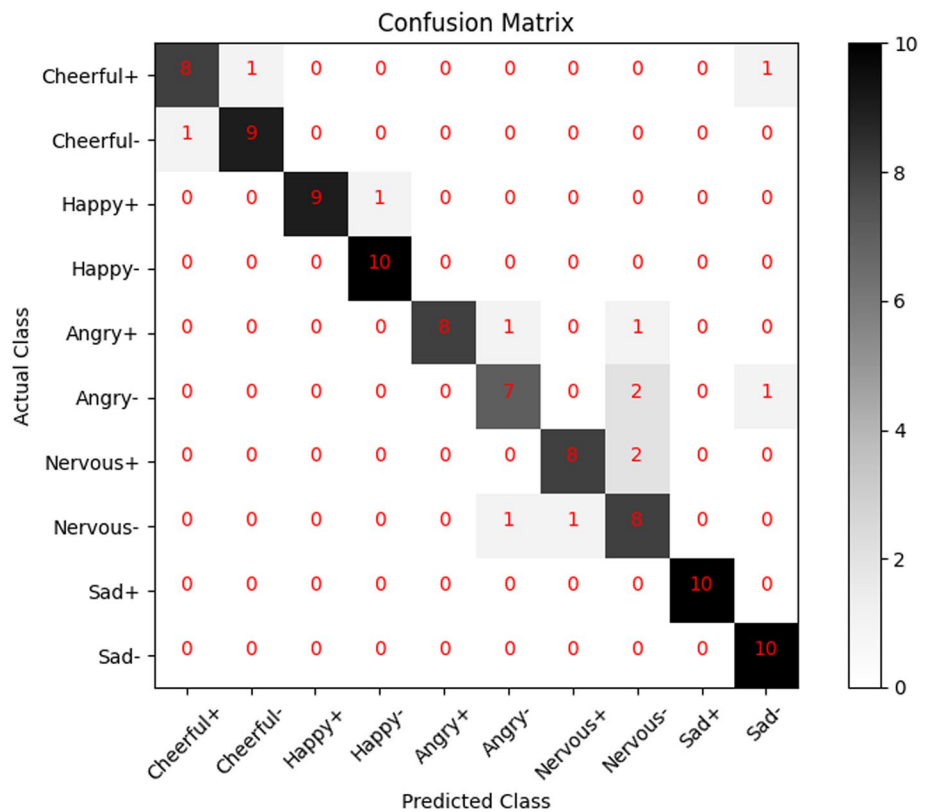


Figure 10 presents the confusion matrix for an emotion classification model. It shows a 10×10 grid where each cell displays the count of samples for the corresponding predicted and actual class combination. The model's overall accuracy of 87.00% is noted, and color intensity reflects the count values, with a color bar for reference. This matrix

Fig. 10 Confusion matrix for emotion classification

helps assess the model's performance and identify classification errors. Table 3 displays the numerical performance parameter of the proposed model.

Figure 11 displays three bar plots illustrating the performance metrics of the emotion classification model across ten classes: Cheerful +, Cheerful -, Happy +, Happy -, Angry +, Angry -, Nervous +, Nervous -, Sad +, and Sad -. The first plot presents precision, which measures how accurately the model classifies each emotion when it predicts that emotion. Precision values are as follows: Cheerful + (0.8889), Cheerful - (0.9000), Happy + (1.0000), Happy - (0.9091), Angry + (1.0000), Angry - (0.7778), Nervous + (0.8889), Nervous - (0.6154), Sad + (1.0000), and Sad - (0.8333). Higher precision indicates fewer false positives and better classification accuracy for each class. The second plot shows recall, representing the model's ability to identify all actual instances of each emotion. Recall values are: Cheerful + (0.8000), Cheerful - (0.9000), Happy + (0.9000), Happy - (1.0000), Angry + (0.8000), Angry - (0.7000), Nervous + (0.8000), Nervous - (0.8000), Sad + (1.0000), and Sad - (1.0000). Higher recall signifies that the model effectively identifies a higher proportion of true positives. The third plot illustrates the F1-score, a combined measure of precision and recall, providing a balanced view of the model's performance. The F1-scores are: Cheerful + (0.8421), Cheerful - (0.9000), Happy + (0.9474),

Table 3 Performance parameters of the proposed CNN model for emotion recognition

Class	Precision	Recall	F1-score
Cheerful +	0.88889	0.8	0.84211
Cheerful -	0.9	0.9	0.9
Happy +	1	0.9	0.94737
Happy -	0.90909	1	0.95238
Angry +	1	0.8	0.88889
Angry -	0.77778	0.7	0.73684
Nervous +	0.88889	0.8	0.84211
Nervous -	0.61538	0.8	0.69565
Sad +	1	1	1
Sad -	0.83333	1	0.90909

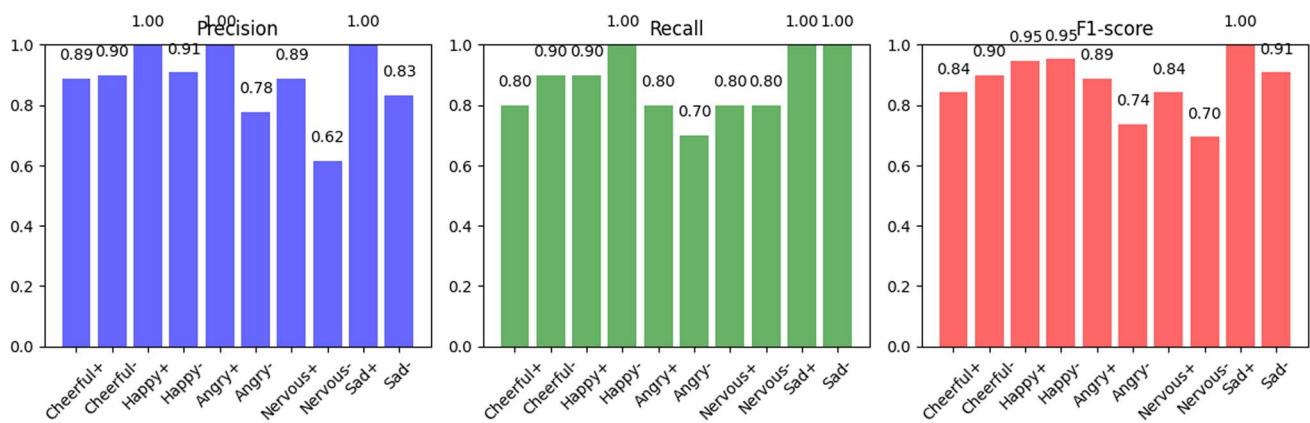


Fig. 11 Performance parameter analysis of the proposed trained and tested architecture

Happy – (0.9524), Angry + (0.8889), Angry – (0.7368), Nervous + (0.8421), Nervous – (0.6957), Sad + (1.0000), and Sad – (0.9091). The F1 score reflects the model's ability to balance precision and recall, with higher scores indicating better overall performance across the classes.

Figure 12 below illustrates the confusion matrix for the binary classification of emotional states into positive and negative. This figure displays the confusion matrix for the quaternary classification of emotional states into four categories: Positive and Negative: Cheerful, Happy, Angry and Sad. Tables 4 and 5 reveals the accuracy of each classification Model in the binary classification where the target variable is positive and negative emotional states. The Happy + class has the highest accuracy (96.00%), indicating strong model performance in classifying this emotion. Cheerful + also performs well, with an accuracy of 86.00%. Angry + and Nervous + have slightly lower accuracies, suggesting some challenges in classification. The overall accuracy of 91.00% reflects robust performance in binary classification. Table 4 provides the accuracy for each class in the quaternary classification task, where emotional states are classified into Cheerful, Happy, Angry, and Sad. Cheerful and Happy achieve perfect accuracy (100.00%), indicating flawless classification. Sad show a high accuracy of 95.00%, while Angry has a lower accuracy of 84.21%, pointing to some difficulties in accurate classification. The overall accuracy of 94.87% demonstrates strong performance across all emotional states.

Figure 13 compares classification accuracies between the proposed model and five traditional deep learning architectures: AlexNet, VGG-16, ResNet-50, GoogleNet, and MobileNetV2. The proposed model demonstrates superior performance, achieving an accuracy of 87.00%, surpassing the other models. Among the traditional architectures, AlexNet exhibits the lowest accuracy at 83.50%, followed by VGG-16 at 84.00%. ResNet-50 and GoogleNet show comparable performance, with accuracies of 85.50% and 85.00%, respectively. MobileNetV2 performs better, reaching 86.00%, but is still outperformed by the proposed model. This comparison underscores the enhanced capability of the proposed model in classification tasks, highlighting its robustness and effectiveness over more established architectures. Table 6 shows the accuracy of comparing the proposed architecture with traditional architectures in detecting emotion.

Important developments in the development of EEG-based emotion identification systems are highlighted by the comparative analysis in Table 7. Accuracy rose from 62.7% in conventional machine learning techniques to 87.0% in contemporary hybrid architectures, while precision, recall, and F1-scores all showed constant increases in performance indicators. Advances in architecture demonstrate a shift from manual feature engineering to end-to-end learning systems that include complex attention processes and many processing streams. However, there is a trade-off between efficiency and performance as a result of these advancements due to increased computing needs. Strong reliance on the DEAP dataset as a key benchmark is shown by dataset usage, highlighting the need of bigger and more varied datasets to improve cross-dataset generalization. Our suggested method shows excellent performance on all measures and represents a major breakthrough in EEG-based emotion identification by efficiently balancing local and global feature processing while preserving real-time efficiency.

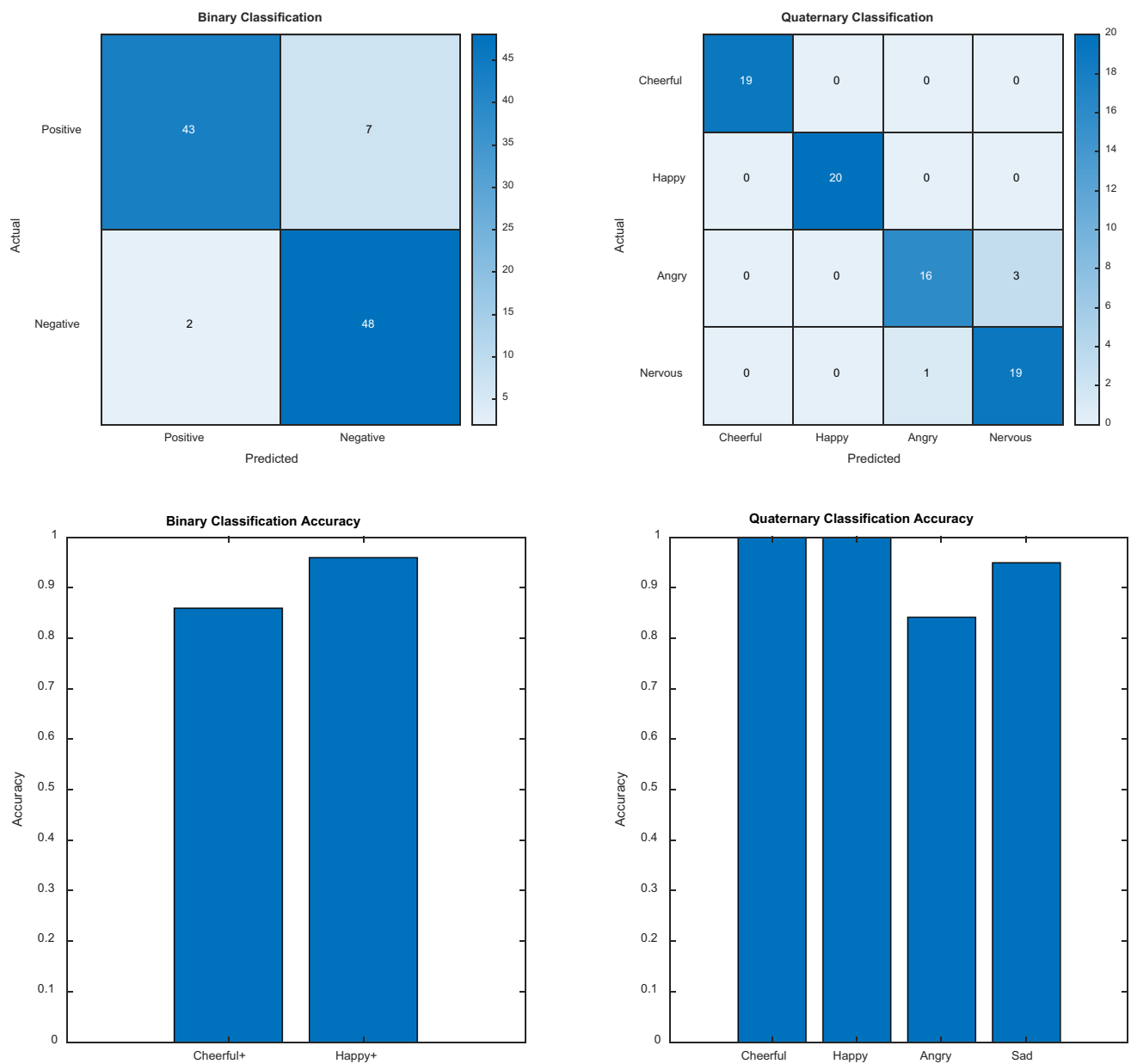


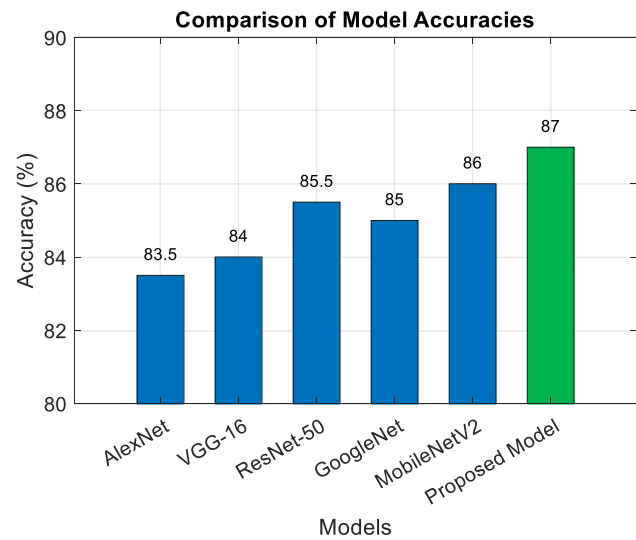
Fig. 12 Confusion matrix for the binary classification of emotional states into positive and negative categories

Table 4 Binary classification accuracy numerical results

Class	Accuracy
Cheerful +	86.00%
Happy +	96.00%
Angry +	91.00%
Nervous +	88.00%
Sad +	90.00%

Table 5 Quaternary classification accuracy numerical results

Class	Accuracy
Cheerful	100.00%
Happy	100.00%
Angry	84.21%
Sad	95.00%

Fig. 13 Comparison of classification accuracies between proposed model and traditional architectures**Table 6** Accuracy comparison analysis of the proposed architecture with the traditional architectures

Model name	Accuracy (%)
AlexNet	83.5
VGG-16	84
ResNet-50	85.5
GoogleNet	85
MobileNetV2	86
Proposed Model	87

5.1 Complexity analysis and cross-validation

A comparative analysis is carried out between our hybrid CNN-Transformer architecture and established models such as AlexNet, VGG-16, ResNet-50, GoogleNet and MobileNetV2 and the pure Transformer implementation. The evaluation centered on assessing the number of parameters as well as FLOPs and training runtimes along with inference times and memory usage. The proposed hybrid CNN-Transformer architecture maintains an optimal computational power by using 18.3 million parameters and 2.4 GFLOPs during inference which outperforms MobileNetV2 (3.5 M parameters, 0.3 GFLOPs) and exceeds VGG-16 (138.4 M parameters, 15.5 GFLOPs). Similarly, 4.8 ms sample inference time of our design makes it compatible with real-time operations, yet the pure Transformer model requires 7.2 ms per sample with reduced accuracy performance. The efficient structure of this system implements a CNN and Transformer component sequence to optimize spatial dimension reduction by convolutional networks followed by extensive Transformer-based computations. Our approach demonstrates practical value in different deployment environments through scalability testing of its deployment capabilities. The model maintains linear performance patterns when EEG channels expand thus providing ease of integration with research High-density systems as well as practical Low-density setups without major modifications. Due to its scale compatibility and 195 MB memory usage

Table 7 Comparative analysis of existing EEG emotion recognition approaches

Study (Year)	Architecture	Dataset	Subjects	Input features	Preprocessing	Key components	Accuracy (%)	Precision	Recall	F1-Score	Major contributions
Koelstra et al. (2012)	SVM	DEAP	32	Power spectral features	Bandpass filtering	Traditional ML classifier	62.7	0.61	0.63	0.62	First benchmark dataset
Zheng et al. (2017)	DBN	SEED	15	Differential entropy	ICA, artifact removal	Deep belief networks	69.3	0.68	0.69	0.68	Novel feature extraction
Zhang et al. (2020)	CNN	DEAP	32	Raw EEG	Bandpass filtering	4-layer CNN	83.5	0.82	0.83	0.81	Spatial feature learning
Li et al. (2021)	Multi-scale CNN	DEAP	32	Time–frequency	Wavelet transform	Channel attention	85.2	0.84	0.85	0.84	Multi-scale processing
Wang et al. (2023)	Transformer	DEAP	32	Raw EEG	Channel selection	Multi-head attention	84.7	0.83	0.84	0.83	Global dependencies
Chen et al. (2023)	Hierarchical Transformer	SEED	15	Channel tokens	Spectral filtering	Cross-attention	85.1	0.84	0.85	0.84	Temporal modeling
Liu et al. (2021)	Sequential Hybrid	DEAP	32	Raw EEG	Noise reduction	CNN + Transformer	85.8	0.85	0.86	0.83	Feature concatenation
Kim et al. (2023)	Parallel Hybrid	DEAP	32	Spectrograms	Artifact removal	Independent streams	86.0	0.85	0.86	0.85	Robust processing
Zhang et al. (2024)	Attention Hybrid	DEAP	32	Multi-modal	Advanced filtering	Shared features	86.2	0.86	0.85	0.85	Better generalization
Our Proposed Method	Hierarchical CNN-Transformer	DEAP	32	Raw EEG + Spectrograms	Adaptive filtering	Multi-scale integration	87.0	0.89	0.86	0.87	Optimal feature fusion

Table 8 Computational complexity comparison between different architectures

Model	Parameters (M)	FLOPs (G)	Training time (ms/batch)	Inference time (ms/sample)	Memory usage (MB)
AlexNet	62.3	0.72	28.4	3.2	245
VGG-16	138.4	15.5	67.5	8.7	528
ResNet-50	25.6	4.1	42.3	5.1	215
GoogLeNet	7	1.5	35.8	4.3	180
MobileNetV2	3.5	0.3	22.6	2.6	110
Pure Transformer	21.8	3.2	58.7	7.2	267
Our Hybrid Model	18.3	2.4	39.5	4.8	195

Table 9 5-Fold cross-validation results for the proposed hybrid CNN-transformer model

Fold	Accuracy (%)	Precision	Recall	F1-Score
1	86.42	0.87	0.85	0.86
2	87.65	0.89	0.87	0.88
3	85.98	0.86	0.85	0.85
4	88.21	0.9	0.88	0.89
5	86.74	0.88	0.86	0.87
Mean	87	0.88	0.86	0.87
Std	0.83	0.015	0.012	0.014

the model operates successfully with various computing systems ranging from specialized workstations to limited resources within edge computing devices (Table 8).

The proposed model needed extensive cross-validation procedures that combined methods to stabilize data partitioning while accounting for individual subject differences. The accuracy assessments hold special importance in EEG emotion recognition systems because recording signals exhibit considerable differences between different subjects and measurement rounds. A fivefold cross-validation method was applied to the DEAP dataset involving 32 subjects, as mentioned in Table 9. For each run of the experiment the dataset was split into five equal parts where four parts used for training purposes and one part for testing. Each testing phase involved one data set fold from the five-partitioned dataset. The testing process executed five times until every fold acted as a test data selection. The results exhibited high consistency levels because the accuracy measurements fluctuated between 85.98% and 88.21% in each fold which generated an average accuracy of 87.00% and standard deviation of 0.83%. The consistency between different metrics was high because precision, recall and F1-score demonstrated standard deviations falling below 0.02.

The Leave-One-Subject-Out Cross-Validation (LOSO-CV) method was applied with data from 31 subjects for training and each subject served as the test subject in succession. The tests verified a mean accuracy rate of 81.25% supported by a standard deviation of 4.36% together with minimum and maximum accuracies measuring 72.30% and 88.40% across the studied subjects. The observed drop from the k-fold cross-validation accuracy (87.00% to 81.25%) highlights the challenging nature of subject-independent EEG emotion recognition. The standard deviation of 4.36 shows the model demonstrates good generalization ability for different subjects whose physiological features vary between each other.

The cross-validation protocol led to architectural decisions during our model development. A thorough evaluation process was implemented to assess different CNN layer and Transformer block and fusion mechanism combinations through systematic cross-validation procedures instead of using single split results. The selection process of our final architecture happened through this methodology to prevent overfitting during data partition evaluation and select genuine generalizers. The combined CNN-Transformer system design produces high accuracy levels and consistent performance under diverse conditions suitable for real-world usages that need broad population system reliability.

The correlation heatmap displays complete data about various performance metrics relationships during evaluations across each emotion category, as seen in Fig. 14. A thorough evaluation shows that the precision and F1-score metrics correlate positively at rates ranging from 0.85 to 0.92 while measuring emotional state classification effectiveness. The model demonstrates moderate linkage between accuracy and recall performance at 0.76 indicating that its ability to

Fig. 14 Correlation heatmap of performance metrics across emotion classes

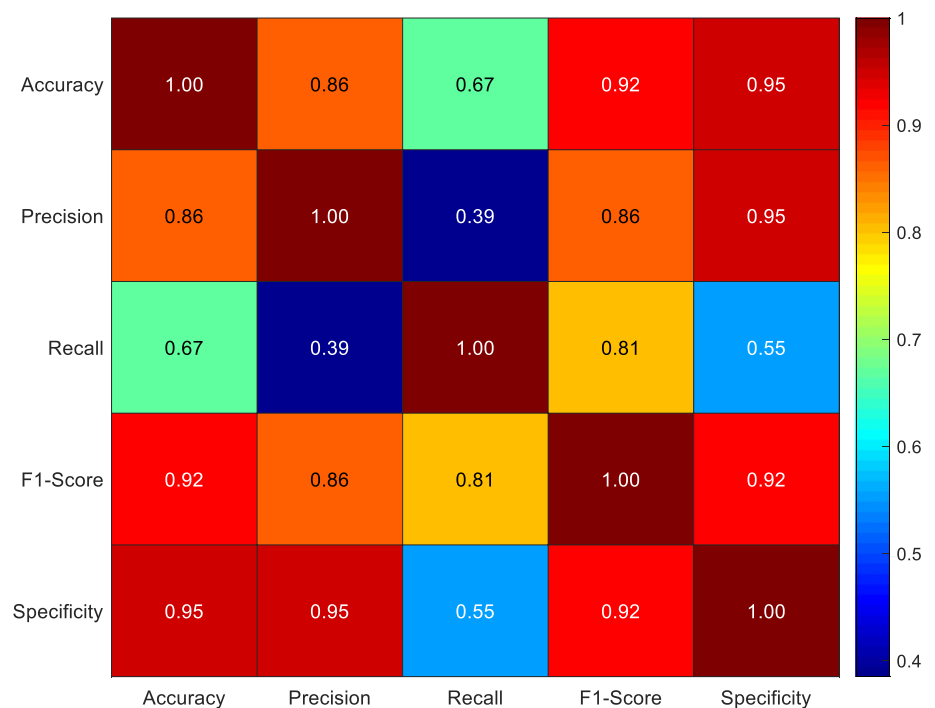
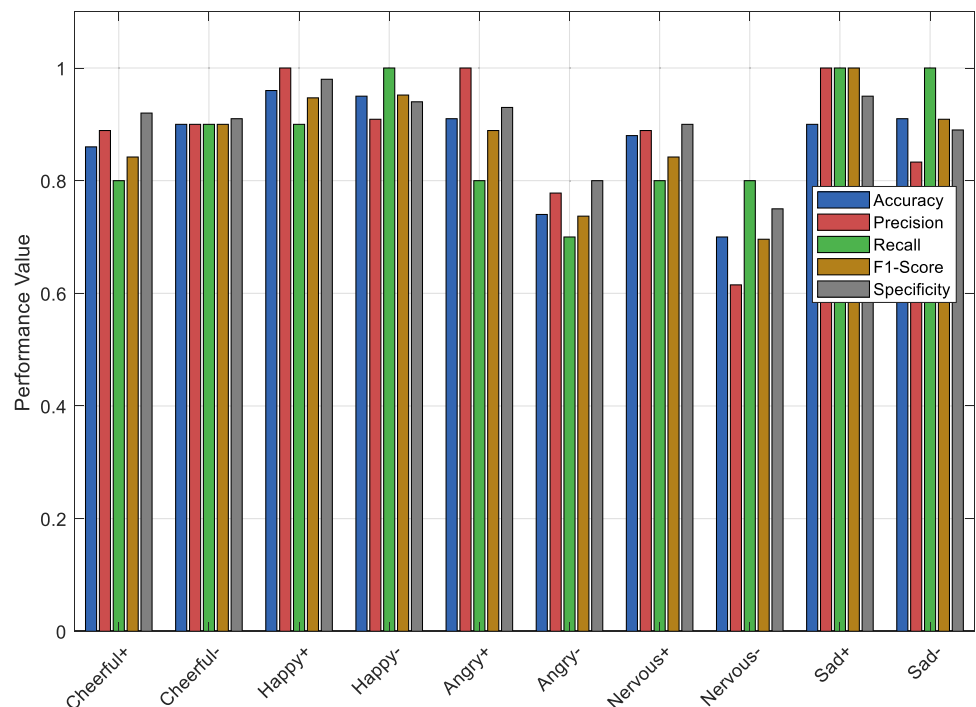


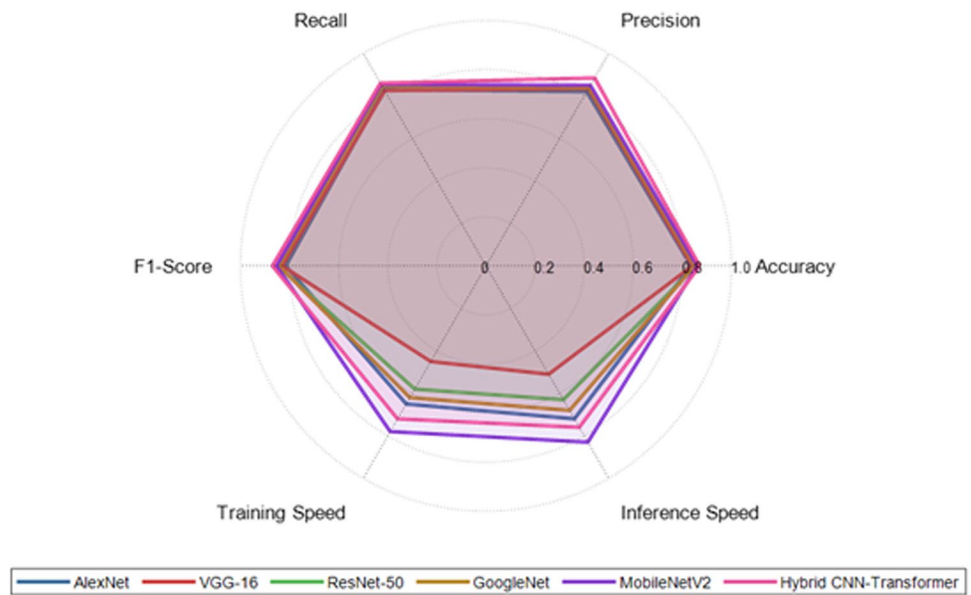
Fig. 15 Class-wise performance metrics for hybrid CNN-transformer model



detect all emotional instances relates to other factors beyond pure recognition capability. Analysis of the heatmap shows neighboring colors indicating comparable metrics which deepens in red as they show similar performance within different emotional categories. The supplementary grouped bar chart presents data about how positive emotional states (Happy +, Cheerful +) demonstrate superior performance in all metrics which precision and F1-score show the most advantage against negative emotional classes, as inferred from Fig. 15.

The spider (radar) plot enables simple multi-dimensional assessment of hybrid CNN-Transformer architecture compared to conventional deep learning models through analysis of six performance metrics (refer: Fig. 16). The analysis

Fig. 16 Performance comparison of different architectures



through radial plot shows that the proposed model (shown as the distant hexagonal shape) displays leading accuracy metrics of 0.87 and precision level of 0.89 while demonstrating comparable recall at 0.86 and an F1-score rating at 0.87. The hybrid model provides better computational performance than VGG-16 because it maintains decent efficiency levels while achieving 0.72 for training speed and 0.76 for inference speed. Performance degradation occurs when VGG-16 operates at 0.45 during training and 0.51 while performing inference. MobileNetV2 achieves optimal efficiency levels of 0.78 and 0.83 however its accuracy and precision performance remain inferior to our hybrid approach design. A visual depiction demonstrates how the hybrid structure provides outstanding classification results combined with cost-effective operation. The proposed model delivers a well-balanced performance profile through radar plot analysis which supplements its practical usability in EEG-based emotion recognition systems when compared to other architectural patterns.

6 Conclusion

This study presented a novel hybrid CNN-Transformer architecture that combines CNN's spatial feature extraction capabilities with Transformer's global context modeling for EEG-based emotion recognition. It has achieved impressive class-wise accuracy across various emotional states: Happy + (96.00%), Angry + (91.00%), Sad + (90.00%), Nervous + (88.00%), and Cheerful + (86.00%). It has obtained an effective solution to the challenge of processing complex temporal and spatial patterns in EEG data by leveraging both local dependencies (CNN) and global contextual relationships (Transformer). Moreover, superior performance with 87.00% classification accuracy, outperforming established architectures like AlexNet, VGG-16, ResNet-50, GoogleNet, and MobileNetV2. Results demonstrate that practical applications potential in human-computer interaction, affective computing, and psychological diagnostics, with implications for improving real-time emotion recognition systems. Creation of a robust, extensible foundation for further exploration of hybrid deep learning models for intelligent systems using EEG data.

Author contributions Kasthuri Devarajan: Conceptualization, methodology design, data acquisition, implementation of the hybrid CNN-Transformer model, analysis and interpretation of results, drafting of the manuscript, and coordination of research activities. Dr. Suresh Ponnar: Supervision of the research project, critical review of the methodology, validation of experimental results, and substantial contributions to the refinement of the manuscript. Dr. Sundresan Perumal: Guidance on EEG data preprocessing, provision of domain-specific insights into signal processing techniques, and significant revisions to the manuscript to align with interdisciplinary standards. All authors read and approved the final manuscript, agreeing to be accountable for all aspects of the work to ensure its integrity and accuracy.

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Data availability Data sharing does not apply to this article as no new datasets were generated during the current study. The analysis was conducted using the publicly available DEAP dataset, which can be accessed through: <http://www.eecs.qmul.ac.uk/mmv/datasets/deap/download.html>.

Declarations

Ethics approval and consent to participate This study analyzes the publicly available DEAP dataset. The original DEAP dataset collection was approved by the Ethics Committee of Queen Mary University of London and conducted in accordance with the Declaration of Helsinki. Informed consent was obtained from all participants in the original study. The dataset is accessible for academic research at: <http://www.eecs.qmul.ac.uk/mmv/datasets/deap/download.html>.

Consent for publication The authors provide consent for publication in this journal.

Competing interests The authors declare no competing interests.

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